

LECTURE 1: COMPARISON OF BEV, HEV, FCEV AND IC ENGINE DRIVE VEHICLES

WHAT IS AN ELECTRIC VEHICLE AND ITS CONSTRUCTION?

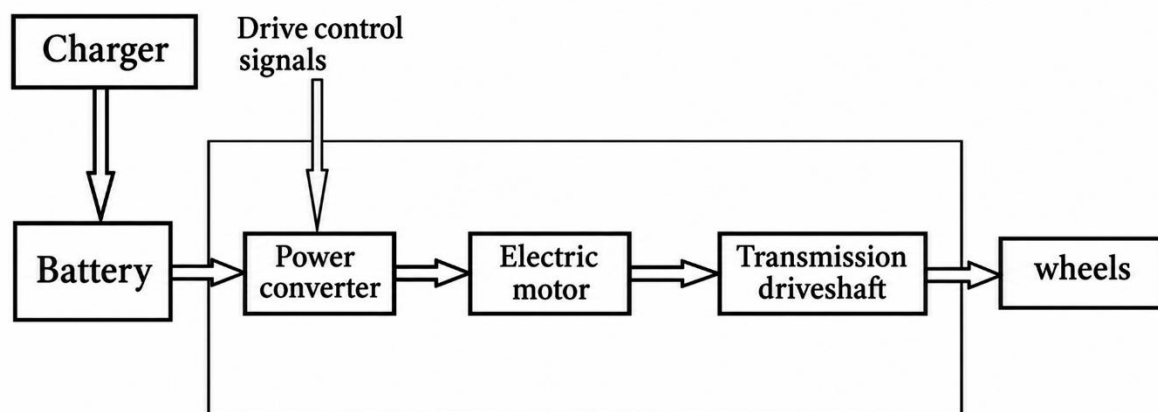
An electric vehicle is a type of vehicle that utilizes one or more electric motors for propulsion instead of relying completely on an internal combustion engine.

Therefore, the rotational mechanical energy required for forwarding the vehicle is produced by electric motors. The required electricity for operating these electric motors is provided by either by battery packs or fuel cells mounted in the vehicle itself.

The block diagram of a typical electric vehicle is shown in the following diagram.

Electric vehicles operate by converting electrical energy stored in batteries into mechanical energy through electric motors, enabling propulsion while offering energy efficiency and zero tailpipe emissions.

The different components of the electric vehicle and their functions are described below



Energy storage

Electric vehicles store energy in battery packs, typically composed of lithium-ion cells due to their high energy density and long cycle life. These batteries supply direct current (DC) electricity to the vehicle's systems and propulsion components, forming the core energy source for EV operation. In practice, the battery packs used in electric vehicles are made up of a number of lithium-ion cells. The distance range per charge of an electric vehicle is determined from the capacity of the battery pack used.

Electric Motors

Instead of an internal combustion engine, EVs use **electric motors** to generate motion. When electricity flows through the motor, it creates a **magnetic field**, causing the rotor to spin. This rotational motion is transferred to the wheels, propelling the vehicle forward. EVs may use **AC or DC motors**, with AC motors often paired with inverters for precise speed and torque control.

Electronic Controller

It is an electronic device used for controlling and monitoring the operation of control system of the electric vehicle. The primary function of the electric vehicles controller is to regulate the flow of electric energy from the battery pack to the electric motors in response to the pressure applied on the accelerator paddle. It also control speed and torque with regenerative braking.

Braking System

In electric vehicles, the regenerative braking system is commonly used. In this braking system, the electric motor act as a generator and converts the mechanical energy back into electrical energy, when brakes are applied. This regenerated electrical energy is stored in the battery packs. This helps in increasing the distance range and overall efficiency of the vehicle.

Battery Charger

In an electric vehicle, an onboard battery charger is provided to recharge the battery pack. This onboard battery charger is plugged into an external source of electricity to charge the vehicle batteries. The onboard charger acts as a rectifier and voltage control device and thus it converts alternating current from the external source into direct current and provide a constant voltage supply to charge the battery pack.

ADVANTAGES OF EVS

1. **Environmental Benefits:** EVs produce zero tailpipe emissions, reducing greenhouse gases and air pollutants such as nitrogen oxides (NOx) and particulate matter, which improves air quality and public health
2. **Lower Running and Maintenance Costs:** Electricity is generally cheaper than gasoline, and EVs have fewer moving parts, eliminating the need for oil changes,

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exhaust repairs, or timing belt replacements, which reduces long-term maintenance costs.

3. **Performance and Driving Experience:** Electric motors provide instant torque, resulting in smooth acceleration and responsive handling. EVs are also quieter, reducing noise pollution and driver fatigue.
4. **Convenience:** EVs can be charged at home, work, or public stations, often overnight, eliminating frequent trips to gas stations. Advanced charging technologies and battery-swapping services further enhance convenience
5. **Energy Security and Sustainability:** EVs reduce dependence on oil imports and can be powered by renewable energy sources, contributing to energy diversity and climate change mitigation
6. **Incentives and Technological Advancements:** Many governments offer grants, rebates, and tax incentives for EV adoption. Technological improvements in batteries and charging infrastructure continue to enhance range and efficiency.

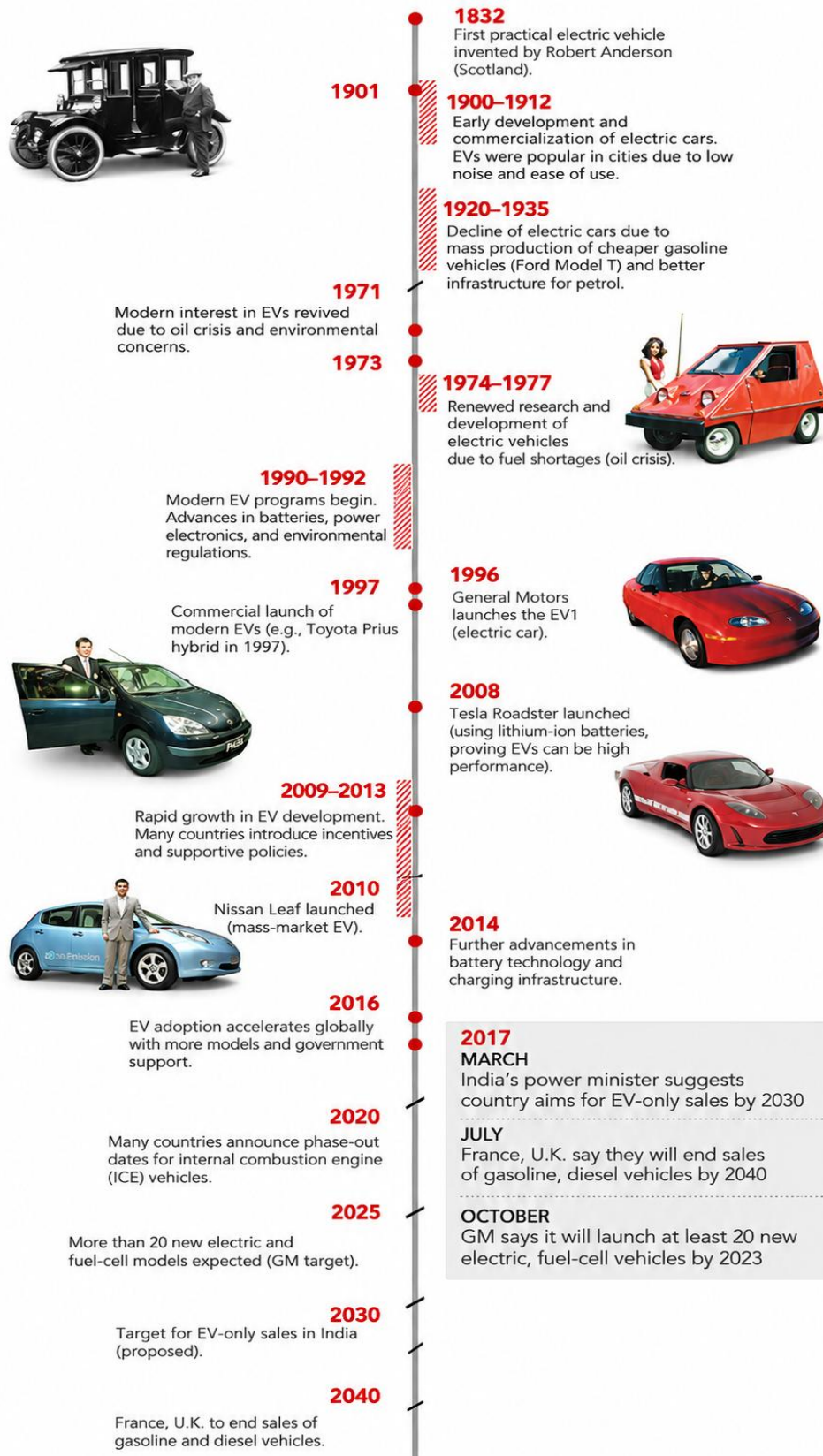
DISADVANTAGES OF EVS

1. **High Upfront Costs:** EVs are generally more expensive than conventional vehicles, primarily due to battery costs
2. **Range Anxiety:** Limited driving range and concerns about running out of charge can deter potential buyers, especially in areas with sparse charging infrastructure
3. **Charging Infrastructure Limitations:** Insufficient public charging stations in some regions make long-distance travel less convenient
4. **Battery Issues:** EV batteries are heavy, expensive to replace, and their production can have environmental impacts. Disposal and recycling of old batteries also pose challenges.
5. **Performance Limitations:** Some EVs may be slower at top speeds compared to high-performance internal combustion engine vehicles, and their weight can affect handling
6. **Limited Vehicle Choice:** Although expanding, the variety of EV models is still smaller than conventional cars, which may limit options for consumers.
7. **Environmental Impact of Manufacturing:** The production of EVs, particularly batteries, can have a higher initial environmental footprint compared to conventional vehicles

COMPARISON OF INTERNAL COMBUSTION ENGINE VEHICLES AND ELECTRIC VEHICLES

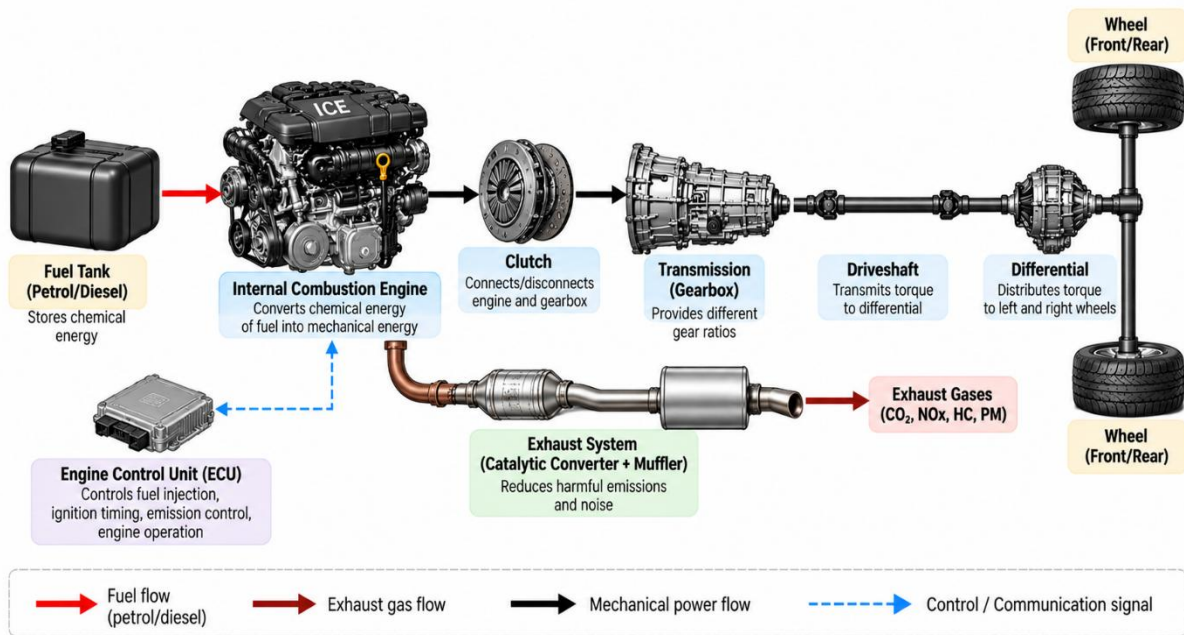
Internal Combustion Engines (ICE)	Electric Vehicles (EV)
More operational cost	Less operational expenses for EVs, i.e., electricity and repair/maintenance
Less capital cost	Higher capital and manufacturing costs bring the total price above that of ICE vehicles
CO ₂ Emission: Petrol – 0.2325 kg/km; Diesel – 0.273 kg/km	EVs have a better emissions profile than internal combustion vehicles. CO ₂ emission – 0.103 kg/km
Engine generates heat that can keep the cabin warm	Range decreases in colder temperatures due to sluggish battery performance and inefficient discharge; motors operate less efficiently
Not a mobile asset	Utilities can use EVs as mobile assets
Gas-powered vehicles take a few minutes to fill their tanks completely	Takes a few hours (0.5 to 8 hours) depending on charger type. Fast chargers take an average of 15–30 minutes to charge the battery
High specific energy (energy per unit mass) of fuel	Low specific energy of battery in EV
Fuels used in ICEVs have high power density	Low power density (power transfer rate per unit volume)
Noisy operation	Quiet operation
Powertrain: IC Engine	Powertrain: Motor
Fuel tank occupies less space and has less weight	Battery occupies more space and is heavy
Braking energy is not recovered	Braking energy can be recovered through regenerative braking
Engine efficiency – approximately 30%	Motor efficiency – approximately 80%
Need to pick up some speed to deliver maximum torque	Produces maximum torque instantly after starting
Uses hydrocarbons	Uses electricity from different resources

EVOLUTION OF ELECTRIC VEHICLES



Internal Combustion Engine (ICE)

An ICE vehicle converts chemical energy stored in liquid hydrocarbons (gasoline or diesel) into thermal energy via controlled internal combustion. Piston movement drives a crankshaft connected to a mechanical multi-speed transmission (manual, automatic, or CVT) that controls torque delivery to the differential and wheels.



Efficiency & Energy Loss: Well-to-wheel thermal efficiency is low (~20%–35%). Over 65% of energy is lost as waste heat through exhaust gas and cooling systems or frictional loss in mechanical gearing.

Emissions: Direct tailpipe output of carbon dioxide (CO_2), nitrogen oxides (NO_x), carbon monoxide (CO), and unburnt hydrocarbons.

Best Use Case: High-speed highway driving and remote long-haul routes where electrical infrastructure is completely absent.

TYPES OF ELECTRIC VEHICLES

Depending on the technology used and needs of applications, there are several different types of electric vehicles. The four most commonly used types of electric vehicles are given below

- Hybrid Electric Vehicle (HEV)
- Plug-in Hybrid Electric Vehicle (PHEV)
- Battery Electric Vehicle (BEV)
- Fuel Cell Electric Vehicle (FCEV)

Let us now discuss about each of these types of electric vehicles in detail along with their construction, working, and features.

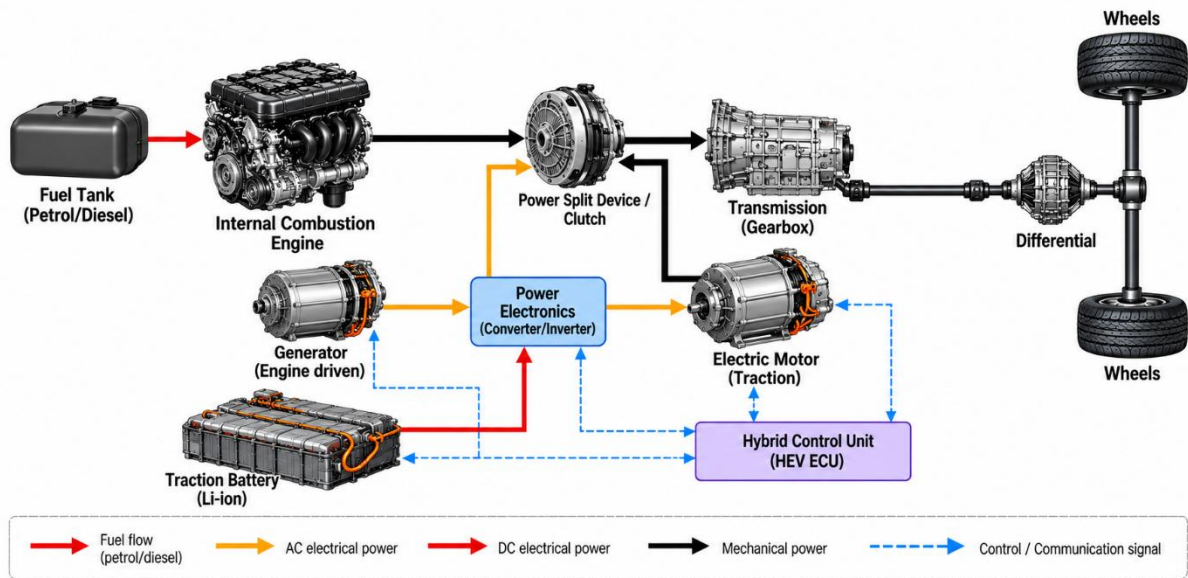
Hybrid Electric Vehicle (HEV)

An HEV fuses an internal combustion engine with an electric motor and a small high-voltage traction battery pack (typically 1–2 kWh). It cannot be plugged into the power grid; battery energy is generated onboard via regenerative braking (converting braking kinetic energy back into electricity) and by using the engine as a generator.

Therefore, in hybrid electric vehicles, both fossil fuels and electricity are the source of energy. The ICE and electric motors are operated in a manner that they can optimize the efficiency and performance of the vehicle.

Overall, a hybrid electric vehicle combines the benefits of both ICE vehicle and all-electric vehicle. Thus, these vehicles are mainly known for their high fuel efficiency.

The block diagram of a typical hybrid electric vehicle is shown in the following figure.



The complete working of a typical hybrid electric vehicle in different conditions is explained below

Low-Speed Driving Condition In this condition, the hybrid electric vehicle is propelled by the electric motors powered by the electricity stored in the battery pack. It is because at low driving speeds, a low acceleration is required that can be easily provided by the electric motors more efficiently. Thus, in the low-speed operating conditions, the IC engine is not engaged in the propulsion of the hybrid electric vehicle.

High-Speed and Heavy Acceleration Driving Condition This condition arises when the vehicle climb a slope or hill or runs on a highway or expressway. In this condition, a large amount power to pull the vehicle is required. Thus, both the internal combustion engine and the electric motors operate together and provide sufficient power required for the propulsion. In this condition, the main power is provided by the internal combustion engine and the power produced by the electric motors supplement it.

Braking and Deceleration Condition In this operating condition, the electric motors behave like the generators and convert the motion energy into electricity which is then stored in the battery pack for later use.

The main features and characteristics of the hybrid electric vehicles are listed below

- Hybrid electric vehicles are better in terms of performance and fuel efficiency.

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- Hybrid electric vehicles do not use IC engine all the times. Thus, they also help in reducing the greenhouse gas emission and environment pollution.
- Hybrid electric vehicles provide a smooth acceleration under different driving conditions. This is accomplished with the help of a complex control mechanism.
- Hybrid electric vehicles have a longer driving range.
- Hybrid electric vehicles do not need to be recharged on a charging station. Instead, they are recharged by IC engine and electric motors during operation.

Efficiency & Energy Loss: Thermal efficiency reaches ~35%–45% due to engine load optimization and recapture of city braking energy.

Best Use Case: Urban stop-and-go commuting without relying on external charging infrastr

PLUG-IN HYBRID ELECTRIC VEHICLE (PHEV)

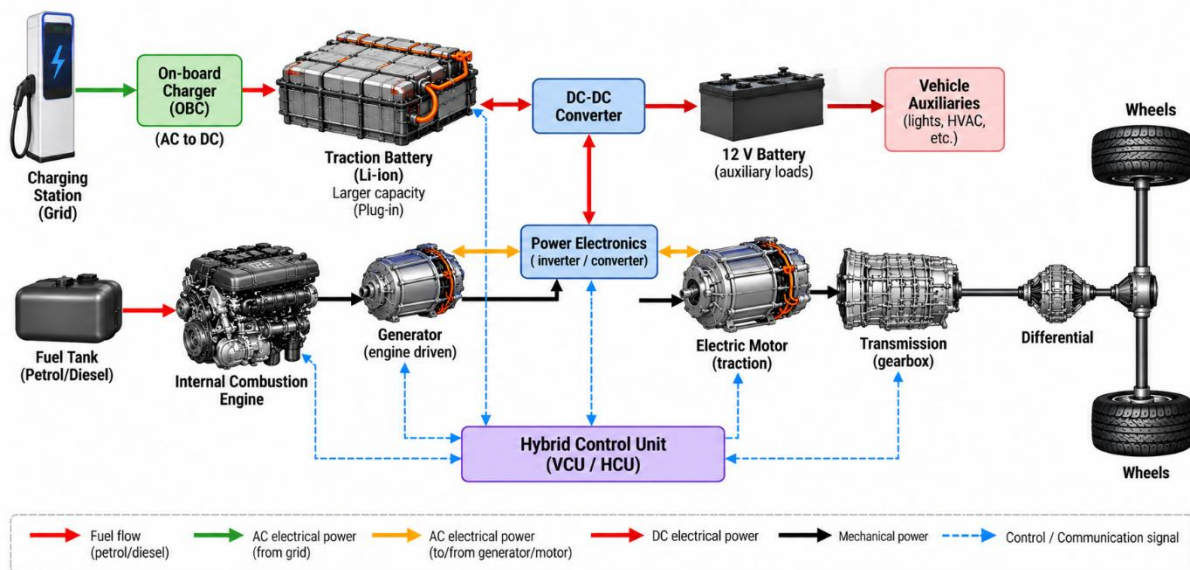
The plug-in hybrid electric vehicle or PHEV is an advanced version of hybrid electric vehicle (HEV). Thus, its construction is almost similar to the hybrid electric vehicle, but it consists of a charger for recharging the battery pack by plugging into an external power supply.

Thus, we can describe a plug-in hybrid electric vehicle as a hybrid electric vehicle with an external charging system.

Another major point that differentiates a plug-in hybrid electric vehicle from a normal hybrid electric vehicle is that it can run in any of the following two modes

- **All electric mode** Only electric motors provide the propulsion power to the vehicle.
- **Hybrid mode** Both IC engine and electric motor pull the vehicle.

The block diagram of a typical plug-in hybrid electric vehicle (PHEV) is shown in the following figure.



The working of a typical plug-in hybrid electric vehicle in different modes is explained in below

All-Electric Mode In the all-electric mode, the PHEV uses the electricity stored in the battery pack to operate the electric motors. These electric motors propel the vehicle forward. In this mode, there is no fuel consumption. Thus, the vehicle operates silently and do not produce any kind of emission.

Hybrid Mode Once the battery pack has been fully discharged, the controller automatically switches the vehicle from all-electric mode to the hybrid mode. In this mode, both the IC engine and the electric motors operate together to propel the vehicle.

Features of Plug-In Hybrid Electric Vehicle

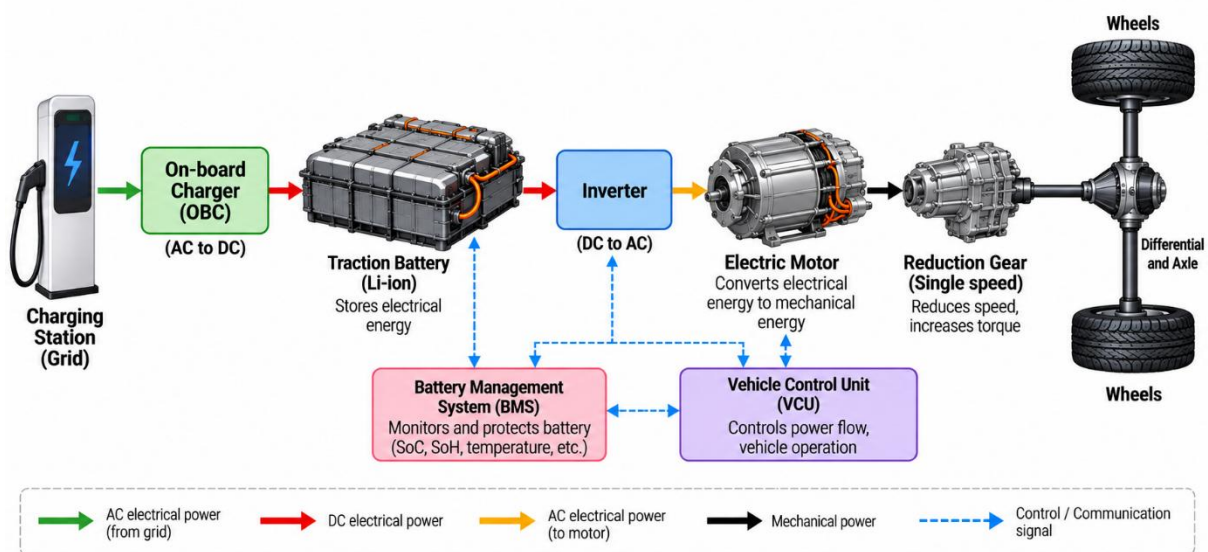
Some of the important features of the plug-in hybrid electric vehicles are listed below

- When PHEVs operate in the all-electric mode, they do not cause noise and pollution.
- PHEVs can operate in hybrid mode. In this mode, they can provide heavy acceleration and high speeds.
- In hybrid mode, the PHEVs can provide longer distance range.
- Regenerative braking system is used in PHEVs that recharges the battery pack by utilizing the kinetic energy of vehicle during braking or deceleration period.
- PHEVs also support external charging.
- PHEVs have higher efficiency. Consequently, they consume less fuel.

BATTERY ELECTRIC VEHICLE (BEV)

A type of electric vehicle in which only electric motors powered by rechargeable batteries are used is called a **battery electric vehicle (BEV)**. Since, in these electric vehicles, there is no other sources of energy like petrol or diesel and their operation completely relies on electricity. Hence, the battery electric vehicles are also known as **All-Electric Vehicles**. An onboard inverter regulates power to permanent-magnet or induction electric motors that drive the axles.

The block diagram of a typical battery electric vehicle is shown in the following figure.



In a battery electric vehicle, the electric energy is supplied from battery pack to the electric motor through an inverter. The inverter converts the direct current electricity of battery into alternating current electricity.

When driver presses the accelerator paddle, an electric signal is sent to the controller of the vehicle. The controller gives instruction to the inverter to regulate the voltage or frequency of the alternating current electricity to adjust the speed of the motor and hence the speed of the vehicle. The motor shaft is connected to wheels of the vehicle directly or through a gearbox.

In case when the brakes are applied, the electric motor acts as a generator and converts the kinetic energy of the moving vehicle back into electricity. This braking technology is called regenerative braking. The regenerated electrical energy is utilized to recharge the battery pack. This is how a battery electric vehicle typically works.

The following some important features of battery electric vehicles (BEVs)

- In battery electric vehicles, there is no combustion of fossil fuels. Thus, they do not pollute the environment.
- The cost of electricity is lower than the cost of fossil fuels like diesel and petrol. Thus, the operating cost of battery electric vehicles is comparatively lower than internal combustion engine vehicles.
- The maintenance cost of battery electric vehicles is also less than ICE vehicle, because they have less number of moving components.
- Electric motors produce less noise when operating as compared to internal combustion engines. Thus, the battery electric vehicles have relatively quiet operation.
- Battery electric vehicles are best option for heavy traffic conditions that require frequent start and stop. This is because they have quick acceleration and deceleration characteristics.
- Battery electric vehicles are more energy efficient, as they use regenerative braking system to recover energy during braking period.

Efficiency & Energy Loss: Tank-to-wheel efficiency ranges between 75%–90%. Minimal thermal loss occurs; torque is delivered instantly from 0 RPM, eliminating multi-speed transmissions.

Emissions: Zero local tailpipe emissions. Lifecycle emissions depend on grid energy mix (renewable vs. coal/gas power plants).

Best Use Case: General consumer travel, commuter routes, and regional transit supported by home or public fast-charging infrastructure.

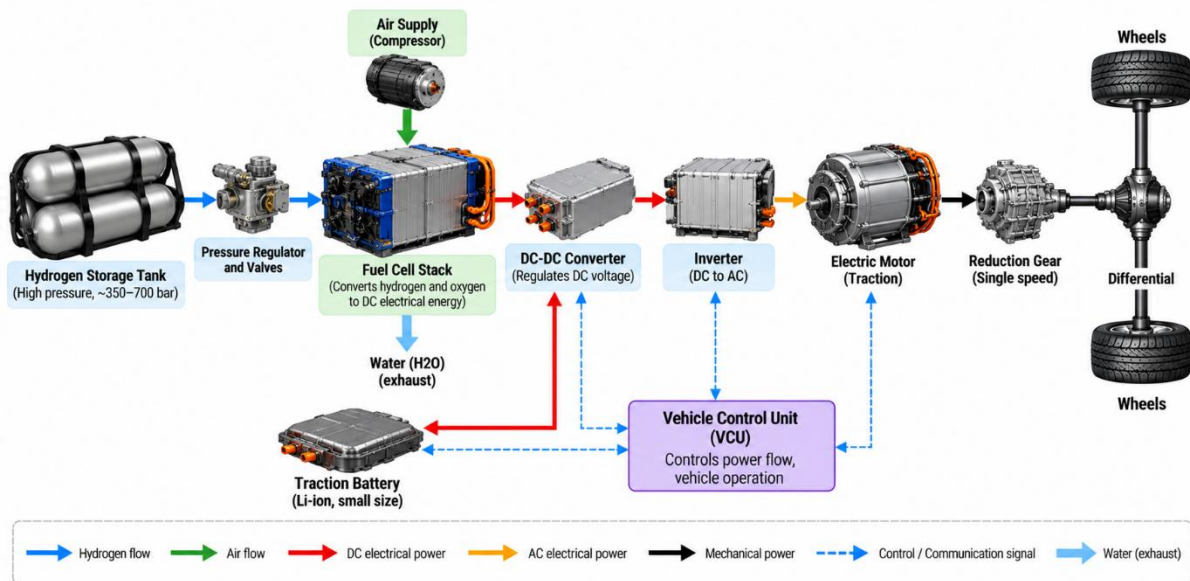
FUEL CELL ELECTRIC VEHICLE (FCEV)

A type of electric vehicle in which fuel cell technology is used for generating electricity to power up the electric motors and propel the vehicle is called a **fuel cell electric vehicle (FCEV)**.

In FCEVs, high pressure hydrogen gas is used as the fuel. It is stored in a leakproof storage tank. This hydrogen gas is fed from the storage tank into fuel cells where it reacts with oxygen to produce electricity and water vapors. This generated electricity is then utilized for operating electric motors.

The fuel cell electric vehicles are also known as zero-emission vehicles because they do produce any kind of emission.

The block diagram of a typical fuel cell electric vehicle is shown in the following figure.



The working of fuel cell electric vehicle is quite straight forward. In these vehicles, the hydrogen gas is fed into a fuel cell stack where it reacts with oxygen of air and produces electricity, heat, and water vapors. The electricity generated is utilized for powering up the electric motors of the vehicle.

The following are some key features and characteristics of fuel cell electric vehicles

- The byproduct of hydrogen gas combustion is water vapors. Thus, the fuel cell electric vehicles cause zero emission and hence do not pollute the environment.

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- Fuel cell electric vehicles have onboard electricity generation mechanism. Thus, they provide longer driving ranges.
- The refueling time of hydrogen gas is almost equal to the refueling time of petrol or diesel, which is very quick.
- The hydrogen gas is used as the fuel in FCEVs which is generally obtained from renewable and low-carbon sources. Hence, these vehicles help in reducing environmental pollution.
- Fuel cell electric vehicle are highly fuel/energy efficient.
- Fuel cell electric vehicles have silent operation.

Efficiency & Energy Loss: Tank-to-wheel efficiency is ~40%–60%. However, well-to-wheel efficiency drops significantly when accounting for hydrogen production (electrolysis), compression, transport, and dispensing.

Emissions: Zero tailpipe pollutants; the sole chemical byproduct is pure water vapor (H_2O).

Best Use Case: Commercial transportation, long-haul heavy trucking, and fleets requiring quick refueling turns without heavy battery weight penalties.

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Feature	ICE	HEV	BEV	FCEV
Primary Energy Source	Petrol / Diesel	Petrol + Battery	Electricity (Grid)	Hydrogen Gas (H_2)
Drivetrain	Engine + Transmission	Engine + Motor + Small Battery	Battery + Electric Motor(s)	Fuel Cell + Motor + Buffer Battery
Tailpipe Emissions	CO_2 , NO_x , particulates	Low to moderate CO_2	Zero	Zero (Water vapor only)
Refueling / Charging Time	3–5 minutes	3–5 minutes (Gas pump)	20–60 mins (DC Fast) / 4–8 hrs (AC)	3–5 minutes
Average Driving Range	500–900 km	700–1,000 km	300–700 km	500–650 km
Energy Efficiency	~20% – 35%	~35% – 45%	~75% – 90%	~40% – 60%
Maintenance Needs	High (frequent oil, filters, spark plugs)	Medium (combines engine & battery maintenance)	Low (fewer moving parts, no oil changes)	Moderate (fuel cell stack monitoring, filters)
Infrastructure Status	Fully mature	Uses existing gas stations	Expanding rapidly	Very sparse/limited